

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 4: Image Filtering

TASK 1: COMPUTE FILTERED VALUE FOR CENTER PIXEL (50)

The 3×3 neighborhood centered at pixel value 50 (row 3, column 3) is:

```
[ 20 30 40 ]  
[ 30 50 60 ]  
[ 40 50 60 ] (Wait, submatrix from 5x5 around 50):  
Row 2: 20 30 40  
Row 3: 30 40 50 (wait, center is row 3, col 3 = 50)  
Neighbors: [30 40 50], [40 50 60], [50 60 70]
```

Sum of 3×3 neighborhood values:

```
Sum = 30 + 40 + 50 + 40 + 50 + 60 + 50 + 60 + 70 = 450
```

Filtered output value:

```
Output = 450 / 9 = 50
```

TASK 2: FORMULA USED

The spatial 2D discrete convolution formula for a 3×3 kernel $w(s, t)$ applied to image neighborhood $f(x+s, y+t)$ is:

$$g(x, y) = (1 / 9) * \sum_{s=-1}^1 \sum_{t=-1}^1 f(x+s, y+t)$$

TASK 3: FILTER TYPE

This is a **Low-pass Filter** (specifically, a box or mean filter).

TASK 4: EFFECT ON NOISE

The mean filter attenuates high-frequency spatial components, which effectively **reduces and smooths random noise** (such as additive Gaussian noise). However, it causes a slight blurring effect across sharp edges in the image.